Algebra

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Part I

Theory

Chapter 1

Complex Numbers

Matrix 1: A line $\overline{a}z+a\overline{z}+c+\overline{c}=0$ is taken in the argand plane. The equation of line after applying the transformation given, is to be found. Column I contains the transformation while column II contains the resulting equations.

Column I	Column II
(P)The line is translated a distance $Re(a)$ along the positive x-axis , $Im(a)$ along positive y-axis , and then reflected in the real axis	(A)$\overline{az}+az=c+\overline{c}+2Re(ia^2)$
(Q)The line is translated a distance $Re(a)$ along the positive x-axis , $Im(a)$ along positive y-axis , and then reflected in imaginary axis	(B)$\overline{az}+az+c+\overline{c}=0$
(P)The line is translated a distance $Im(a)$ along the positive x-axis , $Re(a)$ along negative y-axis , and then reflected in the real axis	(C)$\overline{az}+az+c+\overline{c}-2\left\|a\right\|^2=0$
(P)The line is translated a distance $Im(a)$ along the positive x-axis , $Re(a)$ along positive y-axis , and then reflected in imaginary axis	(D)$\overline{az}+az+2\left\|a\right\|^2=c+\overline{c}$
	(E) $\overline{az}+az+2Im(a^2)=c+\overline{c}$

Chapter 2

Quadratic Equations

2.0.0.1 Linked Comprehension type Problems

Comprehension 1: Let a , b , c be three distinct real numbers and $f(x)$ be a quadratic polynomial satisfying the equation

$\begin{bmatrix} 9a^2 & 9a & 1 \\ 9b^2 & 9b & 1 \\ 9c^2 & 9c & 1 \end{bmatrix} \begin{bmatrix} f(-1) \\ f(-2) \\ f(-3) \end{bmatrix} = \begin{bmatrix} 16a^2+10a \\ 16b^2+10b \\ 16c^2+10c \end{bmatrix}$. Let V be a point of local maxima of $y=f(x)$ and A be the point where $y=f(x)$ meets the x-axis and B be a point on $y=f(x)$ such that AB subtends a right angle at V

Q1: The curve $f(x)$ is given by

A) $f(x) = -\frac{2}{9}x^2 + 2$

B) $f(x) = -\frac{x^2}{10} + 16$

C) $f(x) = -\frac{x^2}{16} + 10$

D) None of these

Q2: The equation of chord AB can be

A) $5x + 6y + 15 = 0$

B) $6x + 5y + 15 = 0$

C) $65x + 27y + 106 = 0$

D) None of these

Q3: The area of the region lying between the curve and the chord AB is

A) $\frac{3445}{8} unit^2$

B) $\frac{3445}{64} unit^2$

C) $\frac{3445}{128} unit^2$

D) None of these

Chapter 3

Probability

3.0.0.1 Matrix Match Type Problems

Matrix 1: A student goes to exam on a bicycle or scooter or car or on foot. The probabilities of his using bicycle, scooter car or being on foot are $\frac{1}{10}$, $\frac{1}{5}$, $\frac{3}{10}$ or $\frac{2}{5}$ respectively. The probability of reaching late at the examination centre by using these modes are $\frac{1}{3}, \frac{1}{4}, \frac{1}{12}$ and $\frac{1}{24}$ respectively. The student reached the exam centre on time. Under Column I, the modes of travel are given and under Column II, the probabilies that the student used the particular mode of transport provided that he reached the exam centre on time. Match the corresponding entries.

Column I	Column II
(P) Bicycle	(A) $\frac{2}{5}$
(Q)Scooter	(B) $\frac{1}{5}$
(R)Car	(C) $\frac{2}{15}$
(S)On Foot	(D) $\frac{4}{15}$

3.0.0.2 Linked Comprehension Type Questions

Comprehension 1: There are n biased coins and n boxes numbered 1 to n. The coins are tossed and each put into its corresponding box. For the k^{th} coin, the odds in favour of appearance of head are k and the odds against are $2n-k$. Let E_n denote the event of selecting the k^{th} box and H denote the event that the selected box contains a Head.

Q1: If $P(E_k) = c \forall k$, then $P\left(E_m|H\right)$ where $m \epsilon \left\{1, 2, 3, ...n\right\}$ is

A) $\frac{m}{n\left(n+1\right)}$

B) $\frac{2m}{n\left(n+1\right)}$

C) $\frac{c}{n+1}$

D) $\dfrac{2}{n+1}$

Q2: If $P\left(E_k\right) \propto k$ for $k = 1, 2, 3........n$. Find $P\left(H\right)$

A) $\dfrac{2n+1}{6n}$
B) $\dfrac{2n+1}{4n}$
C) $\dfrac{2}{3}$
D) $\dfrac{1}{3}$

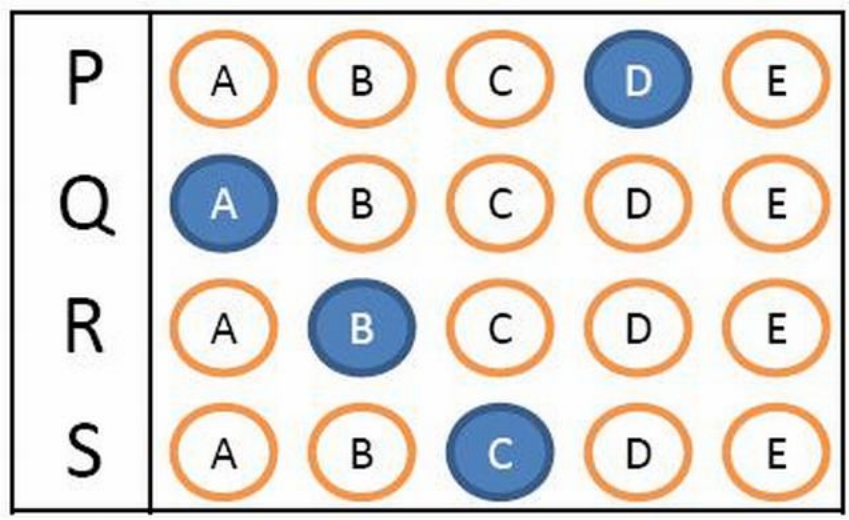
Q3: Evaluate the probability P that only the m^{th} box contains a head .Its value for $m = 3, n = 5$ is

A) 0.12
B) 0.0832
C) 0.648
D) None of these

3.0.0.3 Hints and Solutions

Matrix Match Type Problems

Matrix 1:Answers



Linked Comprehension Type Questions

Comprehension 1 Answers Q1) B Q2) A Q3) C

Chapter 4

Logarithms Worksheet

Q1: Find the domains of definition of the following functions:

a)$\log\left(\dfrac{x^3\left(x^2-4\right)}{\left(x^2-1\right)\left(x+3\right)}\right)$

b)$\sqrt{\log x}$

c) $\sqrt{\log_x(2-x)}$

Q2 Matrix 1: Under Column I , some equations are given . Under Column II, some solutions satisfying some of the equations are given. Match the entry in Column I with the solution satisfying it in Column II.{Note: [ ] is the Greatest Integer Function}

Column I	**Column II**
(P)$\left\|\sin^{-1}x\right\| = \sin^{-1}x + \dfrac{\pi}{6}$	**(A)**-$\dfrac{1}{2}$
(Q)$\ln\|x\| = \|\ln x\|$	**(B)**1
(R)$\|[x]\| = [\|x\|]$	**(C)**$\dfrac{3}{2}$
(S)$\left\|x^2-3x+2\right\| > x^2-3x+2$	**(D)**2

Q3 Comprehension 1: A function $f_n(x)$ is defined for all $n\epsilon N$ and $f_{n+m}(x)$ is defined as $f_{n+m}(x) = f_n(f_m(x))$ where $f_1(x) = \dfrac{2x-1}{x+1}$ for $x\epsilon R - \{-1\}$. Using this definition , $f_2(x) = f_1(f_1(x)) = \dfrac{x-1}{x}$ for $x\epsilon R - \{-1, 0\}$. Similarly $f_3(x) = f_1(f_2(x)) = \dfrac{2-x}{1-2x}$ for $x\epsilon R - \left\{-1, 0, \dfrac{1}{2}\right\}$and so on. Based on the information, answer the questions below.

1: The domain of definition of $g(x) = |\ln(f_{34}(x))|$ is

a) $(-\infty, 1) - \left\{-1, 0, \dfrac{1}{2}\right\}$

b) $(-\infty, 1] - \left\{-1, 0, \dfrac{1}{2}\right\}$

c) $(1, \infty) - \{2\}$

d) None of these

2: The complete solution set of $|f_{71}(x)| > \left|\dfrac{1}{f_{73}(x)}\right|$ is

a) $(-1, 1)$
b) $\mathbb{R} - \{2\} - [-1, 1]$
c) $\mathbb{R} - \{-1, 1, 2\}$
d) None of these

3: The values of x for which $|f_{56}(x)| > f_{56}(x)$ belong to the interval

a) $\left(\dfrac{1}{2}, 2\right)$
b) $(2, \infty)$
c) $(0, 1)$
d) None of these

Q4: (i) Prove that for the logarithmic function $y = \ln|x|$, if the argument takes values in a G.P. , then the corresponding values of the function y are in A.P.

(ii)Prove that for the exponential function $y = e^x$, if the argument takes values in a A.P.. , then the corresponding values of the function y are in G.P.

Chapter 5

Matrices

Q: Invent a 3×3 magic matrix M with the entries 1 , 2 , 3 , ...9. All rows and columns and diagonals add to 15 . The first row could be (8, 3, 4) . What could be M.

Q: If $A = \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}$. Find $A^T A$.

Q: Find the adjoints of the following matrices

a) $\begin{bmatrix} 15 & 7 \\ 8 & 14 \end{bmatrix}$

b) $\begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Q: If $A = \begin{bmatrix} 2x+6 \\ 26 \\ 2z+48 \\ t^2+41 \end{bmatrix}$

$$= \begin{bmatrix} x^2 & -x & 1 \\ y^2 & -y & 2 \\ z^2 & -z & 3 \\ t^2 & -t & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Then , A can be

a) $\begin{bmatrix} 10 \\ 26 \\ 46 \\ 47 \end{bmatrix}$

b) $\begin{bmatrix} 10 \\ 26 \\ 54 \\ 42 \end{bmatrix}$

c) $\begin{bmatrix} 10 \\ 26 \\ 46 \\ 42 \end{bmatrix}$

d) $\begin{bmatrix} 10 \\ 26 \\ 54 \\ 41\frac{9}{25} \end{bmatrix}$

5.0.0.1 Linked Comprehension Type Questions

Comprehension 1: Two vectors $v = (1, 2, 3)$ and $w = (1, 3, 4)$ are taken and put in the columns of a matrix as

$$M = \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}$$

The linear combination of these two vectors can be expressed as

$$c \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + d \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix}$$

Q1: A relation between A , B , c and d is

a) $6c + 8d = A + B$
b) $8c + 6d = A + B$
c) $6c + 8d = 6A + 8B$
d) None of these

Q2: The matrix $\begin{bmatrix} A \\ B \end{bmatrix}$ in the equation

$\begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix}$ is given by

a) $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
b) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$
c) $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$
d) None of these

5.0.0.2 Matrix Match Type Problems

Matrix 1 : Three elementary matrices $E = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$, $G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$ are given. Under column I , resultant matrices (lower triangular) obtained from E, F, G or their inverses are given. Match the matrices given in column I with their corresponding expressions given under column II :

Column I	Column II
(P) $\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$	(A)EFG
(Q) $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}$	(B)$EF^{-1}G$
(R) $\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$	(C)$E^{-1}FG^{-1}$
(S) $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}$	(D)$E^{-1}F^{-1}G^{-1}$
	(E)$G^{-1}F^{-1}E^{-1}$

Part II

System of Linear Equations

Chapter 6

JEE Mains Problems

Q1 : Let α, $\beta \in R$ be such that the system of linear equations

$x + 2y + z = 5$

$2x + y + \alpha z = 5$

$8x + 4y + \beta z = 18$

has no solution. Then $\frac{\beta}{\alpha}$ is equal to :

[JEE Mains 2026 2nd April Shift 1 Q3]

A) -4

B) 4

C) 8

D) -8

{ Soltution : https://www.youtube.com/watch?v=NAct7oeCy2s}

Part III

Quadratic Equations

Chapter 7

JEE Mains Problems

Q1: Let α, $\alpha + 2$, $\alpha \in Z$, be the roots of the quadratic equation
$x(x + 2) + (x +1)(x + 3) + (x + 2)(x + 4) + ...+ (x + n + 1)(x + n - 1) = 4n$ for some $n \in N$. Then $n + \alpha$ is equal to :
[JEE Mains 2nd April 2026 Shift 1 Q1]
A) 0
B) 1
C) 2
D) 3
{Solution: https://www.youtube.com/watch?v=$_W AVSUet9a0$}

Part IV

Probability

Chapter 8

Advanced

Q1: Three students S_1,S_2, and S_3 are given a problem to solve. Consider the following events:

U: At least one of S_1,S_2, and S_3 can solve the problem,

V: S_1 can solve the problem, given that neither S_2 nor S_3 can solve the problem,

W: S_2 can solve the problem and S_3 cannot solve the problem,

T: S_3 can solve the problem.

For any event E, let $P(E)$ denote the probability of E. If

$P(U) = \frac{1}{2}$, $P(V) = \frac{1}{10}$, and $P(W) = \frac{1}{12}$.

then $P(T)$ is equal to :

[JEE Advanced 2025 Paper 1 Q2]

(A) $\frac{13}{36}$

(B) $\frac{1}{3}$

(C) $\frac{19}{60}$

(D) $\frac{1}{4}$

{ Solution : https://www.youtube.com/watch?v=gjzClhgArM4 }

Part V

Matrices and Determinants

Chapter 9

JEE Mains Problems

Q1 : Let $A = \begin{bmatrix} 1 & 2 \\ 1 & \alpha \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 3 \\ \beta & 2 \end{bmatrix}$. If $A^2 - 4A + I = 0$ and $B^2 - 5B - 6I = 0$, then among the two statements :

(S1) : $[(B - A)(B + A)]^T = \begin{bmatrix} 13 & 15 \\ 7 & 10 \end{bmatrix}$

(S2) : $det\,(adj\,(A + B)) = -5$,

[JEE Mains 2nd Apr 2026 Shift 1 Q4]

(A) Only (S1) is correct

(B) Only (S2) is correct

(C) Both (S1) and (S2) are correct

(D) Both (S1) and (S2) are wrong

{Solution : https://www.youtube.com/watch?v=$_{u}TUbsovqU8$}

Part VI

Complex Numbers

Chapter 10

JEE Mains Problems

Q1: Let x and y be real numbers such that $50\left(\frac{2x}{1+3i}-\frac{y}{1-2i}\right)=31+17i$, $i=\sqrt{-1}$.Then the value of $10(x-3y)$ is :

[JEE Mains 2nd Apr 2026 Shift 1 Q2]

(A) 20

(B) 31

(C) 35

(D) 75

{ Solution: https://www.youtube.com/watch?v=7bvmAsRoeiU }

Chapter 11

JEE Advanced Problems

Q1: Let $\mathbb{R}$ denote the set of all real numbers. Let $z1{=}1{+}2i$ and $z2{=}3i$ be two complex numbers, where $i{=}\sqrt{-1}$. Let

$S{=}\{(x,y){\in}\mathbb{R}{\times}\mathbb{R}|x + iy - z_1| = 2|x + iy - z_2|\}$.

Then which of the following statements is (are) TRUE?

[JEE Advanced 2025 Paper 1 Q7]

(A) S is a circle with centre $(-\frac{1}{3}, \frac{10}{3})$

(B) S is a circle with centre $(\frac{1}{3}, \frac{8}{3})$

(C) S is a circle with radius $\frac{\sqrt{2}}{3}$

(D) S is a circle with radius $\frac{2\sqrt{2}}{3}$

{ Solution : https://www.youtube.com/watch?v=cYJl2HBn030 }